S.M. MOHIBUL ABRAR

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SUMMARY

Graduate in Computer Science & Engineering from BRAC University with a curiosity mindset and hands-on experience in Python, data engineering, computer vision, and full-stack development. Skilled in building automated data pipelines, working with AI models, and solving real-world problems through clean, efficient code. Driven by a strong desire to grow through practical, impact-driven engineering work.

SKILLS

Programming Languages: Python (Proficient), JavaScript, C/C++, SQL
AI & Machine Learning: Hugging Face Transformers, Computer Vision, POI Detection, AI Coding Agents
Data Engineering: Web Scraping & Crawling, Data Collection Automation, Data Cleaning & Preprocessing
Data Analysis: Pandas, NumPy, Excel, Google Sheets
Web & Backend: React.js, Node.js, Express.js, RESTful APIs, HTML, CSS
Databases: MySQL, MongoDB – schema design, queries, joins, aggregations
Tools & Platforms: Git, GitHub, Linux (Ubuntu), Bash/Shell Commands
Embedded Systems: Arduino, Raspberry Pi, STM32
Languages: Bangla (Native), English (Proficient)

WORK EXPERIENCE

Data Acquisition Engineer (Intern) **Full-time**
Barikoi Maps, Dhaka *September 2026 – Present*

- Designed and implemented automated data collection pipelines using Python to acquire large-scale geospatial and location datasets efficiently.
- Built web crawlers and scrapers to extract structured data from diverse online sources, handling pagination, rate limiting, and anti-bot measures.
- Queried and managed relational databases to retrieve, filter, and validate acquired datasets for downstream use.
- Performed data cleaning and preprocessing on raw, noisy datasets to ensure quality and consistency across pipelines.
- Developed and applied computer vision techniques for Point of Interest (POI) detection from image data, leveraging pre-trained models from Hugging Face.
- Worked in a Linux (Ubuntu) environment, using shell scripting and command-line tools as part of day-to-day engineering workflows.

EDUCATION

B.Sc. in Computer Science & Engineering **2022 – 2026**
BRAC University, Dhaka *CGPA: 3.34 / 4.00*

Higher Secondary Certificate | Science **2018 – 2020**
Chittagong Cantonment Public College, Chittagong

Secondary School Certificate | Science **2016 – 2018**
Chittagong Government High School, Chittagong

PROJECTS

Bookstore Management System | *React.js, Node.js, Express.js, MongoDB, MySQL*

- Built a full-stack application using the MERN stack with a MySQL relational backend, implementing RESTful APIs for inventory, customer, and sales management.
- Designed database schemas with proper entity relationships and wrote optimized SQL queries for data retrieval and reporting.
- Developed React.js frontend components with clean state management and seamless integration with backend services.

Interactive Data Spreadsheet | *Python, Pandas, Excel, Google Sheets*

- Automated real-world data workflows using Python and spreadsheet tools, incorporating validation logic, transformation pipelines, and dynamic visualizations.
- Applied systematic data cleaning techniques to ensure accuracy, completeness, and consistency of structured datasets.

Dictator Dash | *Python, OpenGL, C++*

- Developed a real-time 2D game in Python using OpenGL, implementing physics-based mechanics including collision detection, projectile motion, and event-driven logic.
- Debugged rendering and physics edge cases through systematic iterative testing and performance profiling.

RELEVANT COURSEWORK

- Artificial Intelligence
- Computer Vision & Image Processing
- Data Structures & Algorithms
- Database Management Systems
- Object-Oriented Programming
- Computer Networks
- Operating Systems
- Robotics (Arduino, RPi, STM32)
- Linear Algebra & Calculus

HONORS & AWARDS

- **Merit-Based Scholarship** | **Highest Distinction in Academics** **2022 – 2023**
  – Awarded by BRAC University for outstanding academic performance in the first year.
- **The Duke of Edinburgh's International Award** | **Bronze Level** **2023**
  – Recognized for leadership, volunteering, fitness, and skill development activities.

EXTRACURRICULAR ACTIVITIES

Convocare Foundation | *President & Founding Member*

– Co-founded and led a student-driven social initiative, coordinating cross-functional teams and overseeing program planning and execution.

CCPC Royal Science Society | *Executive Director*

– Directed science outreach programs and managed organizational operations, fostering analytical and scientific thinking among members.

CGHS Debating Club | *General Member*

– Developed strong communication and logical argumentation skills through competitive debating activities.

REFERENCES

Muhammad Iqbal Hossain, PhD | **Associate Professor, Department of CSE**
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